

Classification for Landslide Detection

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Overview and Objectives

The goal of this competition was to develop a classification model for accurate and reliable landslide detection using multi-source satellite imagery. Landslides, often triggered by natural events such as earthquakes and heavy rainfall, pose significant threats to human life, infrastructure, and the environment. The solution integrates optical data from Sentinel-2's MultiSpectral Instrument (MSI) at Level-2A, which provides 10-meter resolution Red, Green, Blue, and Near-Infrared (NIR) bands, with radar data from Sentinel-1, including VV and VH polarization bands at the same resolution. The radar dataset also includes post-event backscatter and change detection bands, capturing terrain alterations indicative of landslide activity.

Optical imagery offers high visual fidelity, while Synthetic Aperture Radar (SAR) enables consistent monitoring regardless of weather or cloud cover. By combining these complementary data sources, the model is able to detect landslide-affected regions with improved robustness and accuracy, even under challenging observational conditions. The training dataset is based on validated landslide inventories from past earthquake-triggered events.

Data Information

The training and test datasets consist of 12 input channels per image patch. These include four optical bands, Red, Green, Blue, and Near-Infrared (NIR), sourced from Sentinel-2, as well as eight SAR-based channels from Sentinel-1. The SAR channels are divided into ascending and descending orbit data, each containing VV and VH polarization bands along with their corresponding change detection bands (calculated as the difference between post-event and pre-event backscatter). Specifically, the full set of channels is: Red, Green, Blue, NIR, Descending VV, Descending VH, Descending Diff VV, Descending Diff VH, Ascending VV, Ascending VH, Ascending Diff VV, and Ascending Diff VH.

Each image patch is 64×64 pixels in size, with a spatial resolution of approximately 10 meters per pixel. This corresponds to a ground coverage area of 640×640 meters per patch.

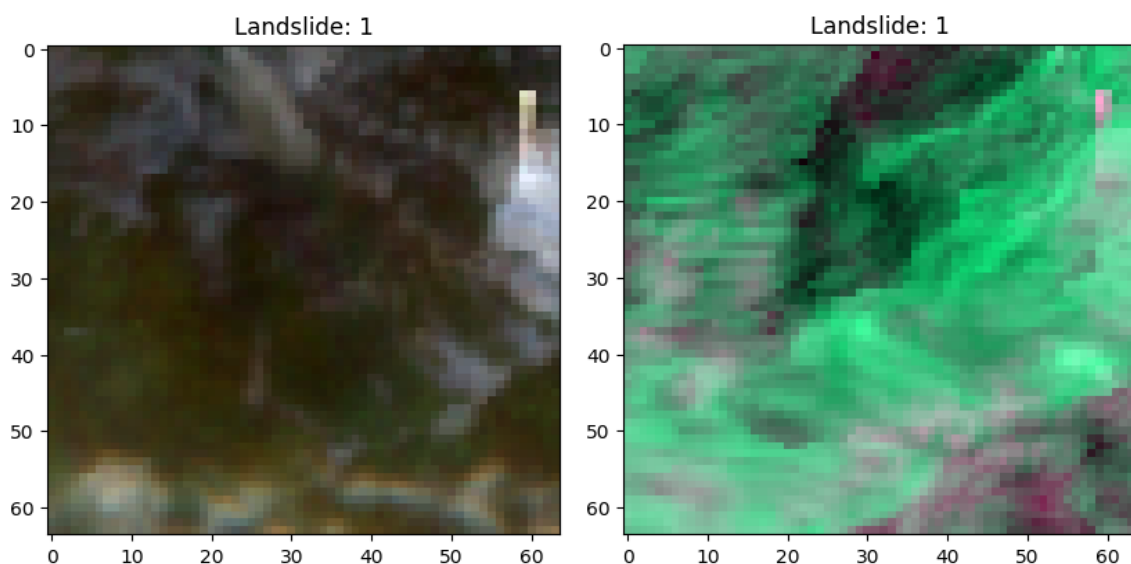


Figure 1: Sentinel-2 RGB image on the left, Sentinel-1 SAR (desc) on the right

System

All experimentation was run on Google Colab pro.

Characteristic	Google Colab pro
Cuda	12.5
Python	3.11.13
Pytorch	2.6
GPU	Tesla T4
RAM	50GB

Solution Description

The final solution is an ensemble of predictions from nine (9) classification models, comprising seven neural networks (NNs) and two gradient boosting models (GBMs). For each neural network, predictions were averaged across all test-time augmentations (TTA = 4, using horizontal and vertical flips) and across five cross-validation folds. In the post-processing stage, a classification threshold of 0.49 was applied to convert predicted probabilities into binary outputs. This threshold was selected based on the highest out-of-fold (OOF) performance.

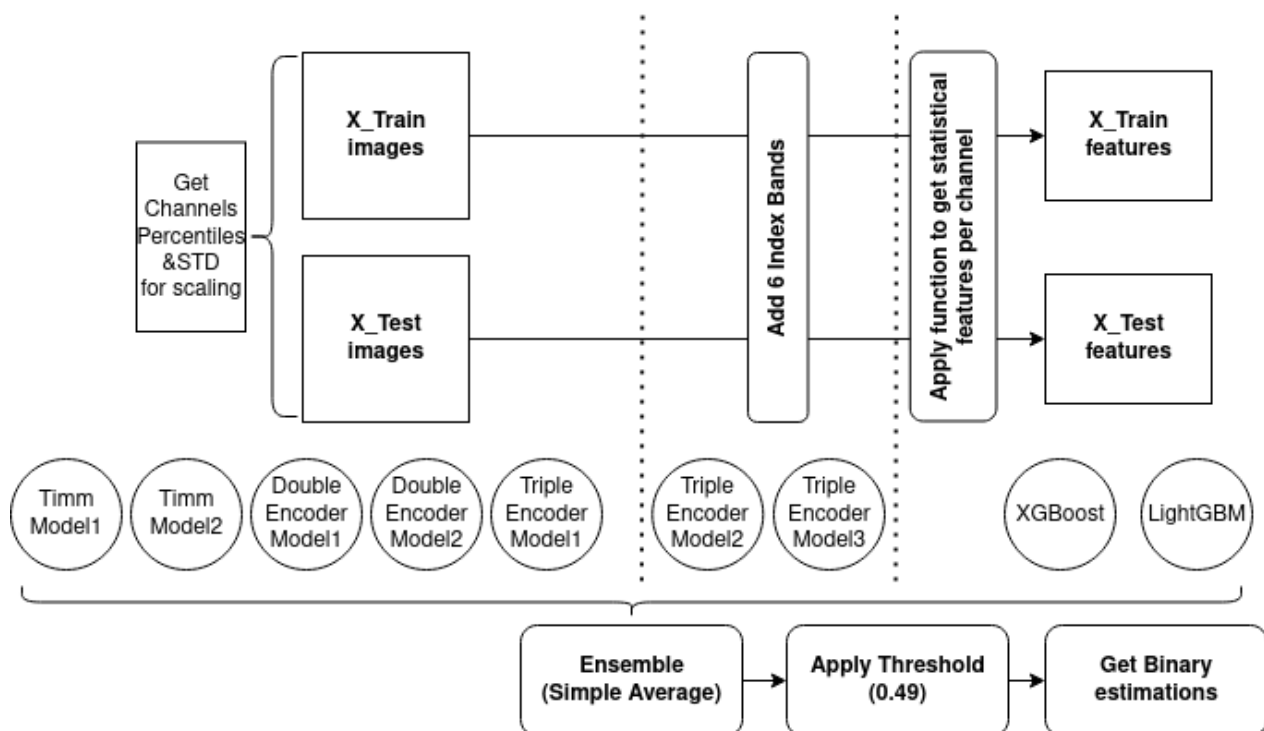


Figure 2: Training and inference FlowChart

Model Architectures and Feature Engineering

For the gradient boosting models (LightGBM and XGBoost), feature-engineered datasets were utilized. These train and test datasets were generated by applying statistical operations, such as minimum, mean, median, maximum, standard deviation, skewness, and kurtosis, independently to each band of every image patch.

For the neural network approach, the PyTorch framework was used in combination with the timm library. Pretrained timm models were either used directly or incorporated into custom architectures. These custom models employed multiple instances of the timm encoder, with each instance dedicated to a specific group of input channels, such as RGBN, SAR, SAR difference (pre/post-event), and additional engineered channels, allowing the model to learn specialized features from each modality.

Data Preprocessing

Two main preprocessing strategies were explored across different experiments, while no preprocessing was applied for the LightGBM and XGBoost models. The two scaling methods used were standard scaling and robust scaling based on the 5th and 95th percentiles to reduce the influence of outliers. In some runs, additional derived channels, such as NDVI and other calculated indices, were added to the original data to enrich the image patches and provide more informative features for model training.

Training Scheme

Neural network training was performed using a custom cosine annealing learning rate schedule, with most models trained for 20 epochs. The loss function used was a simple average of *BCEWithLogitsLoss* and *SmoothF1Loss*, allowing the model to balance probabilistic calibration with F1 score optimization. A 5-fold cross-validation strategy was employed to ensure robust training and to generate reliable out-of-fold (OOF) predictions.

These OOF predictions were later used to fine-tune the ensembling strategy and to determine the optimal classification threshold. The final threshold selection aimed to maximize the F1 score on the OOF data, improving the model's generalization and binary classification performance.

Augmentation

For data augmentation during neural network training, the Albumentations library was used. Augmentations included vertical flips, horizontal flips, and 90-degree rotations, each applied with a 50% probability. During inference, test-time augmentation (TTA) was performed using all four possible flip combinations to improve prediction robustness. Both training and inference for the neural networks were conducted on images resized to 256×256 pixels, whereas the gradient boosting models (LightGBM and XGBoost) operated on the original feature-derived data, requiring no image resizing.

Final Predictions

The final predictions were obtained by taking a simple average of the outputs from the ensemble of models listed in the table below. Finally, a threshold of 0.49 was applied to convert the averaged probability maps into binary masks. This threshold was selected based on its optimal performance on the validation set, maximizing the overall F1 score.

Experimentation Results

Model	Encoder	Channels	Scaling	OOF	pu/pr LB
LightGBM		0-17	None	0.874	0.8792/0.8408
XGBoost		0-17	None	0.8748	0.8786/0.8468
Timm	maxvit_rmlp_tiny_rw_256.sw_in1k	0,1,2,3	Robust	0.8744	0.8854/0.8625
Timm	vit_medium_patch16_reg4_gap_256.sbb_in12k_ft_in1k	4-17	Robust	0.8496	0.8475/0.8524
combinedV2	vit_pwee_patch16_reg1_gap_256.sbb_in1k	0-3 6,7,10,11	Standard	0.8976	0.9148/0.8884
combinedV2	vit_pwee_patch16_reg1_gap_256.sbb_in1k	0-3 6,7,10,11	Robust	0.8963	0.9118/0.8908
combinedV3	vit_pwee_patch16_reg1_gap_256.sbb_in1k	0-3 6,7,10,11 12-17	Standard	0.9003	0.9136/0.8828
combinedV3	caformer_s18.sail_in22k_ft_in1k	0-3 6,7,10,11 12-17	Standard	0.8996	0.9082/0.9001
combinedV3c	vit_medium_patch16_rope_reg1_gap_256.sbb_in1k	0-3 6,7,10,11 4,5,8,9	Standard	0.8943	0.91/0.8956
Ensemble				0.9226	0.9409/0.9058